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Cost-Push vs. Demand-Pull: Evaluating the Predictive Power and Structural Dynamics of Inflation in VAR-Based Models

TEAM 10

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Abstract

By employing multiple enhanced Vector Autoregressive (VAR) methodologies, this paper investigates how cost-push and demand-pull elements influence inflation in the Eurozone. The paper compares traditional Structural VAR (SVAR) models with refined Bayesian SVAR (BSVAR) and Dynamic Factor-Augmented VAR (DFAVAR) techniques using monthly data from January 2015 to January 2024 for 14 macroeconomic indicators. Although traditional VAR and SVAR models are widely used to study inflation dynamics, this approach enables stronger identification and more accurate forecasting of supply-side and demand-side inflation shocks. Furthermore, the study uses long-run restriction techniques and short-run recursive ordering to mitigate estimation uncertainty and overparameterization. While the results indicate that cost-push shocks show more persistent long-run effects, demand-side shocks tend to have a more immediate and pronounced short-run influence on inflation. This finding underlines the necessity of distinct targeted policy responses to different types of shocks. The results suggest that simpler, demand-focused models may improve short-term forecast accuracy. However, incorporating both supply and demand components is essential to capture sustained inflationary trends. The paper advances empirical methodologies for macroeconomic forecasting through refining the VAR frameworks, thereby offering insights that might result in more efficient and targeted policy-making in an evolving economic environment.

1 Introduction

Accurately forecasting inflation remains a significant challenge for economists and policymakers due to its dynamic and evolving nature. Inflation measures the general increases in goods and services prices and is mainly driven by supply and demand forces, also defined as cost-push and demand-pull forces. Cost-push inflation arises from increases in production costs such as wages, raw materials, or energy. It is often the result of supply-side shocks and can lead to stagflation, an environment with rising inflation but stagnant output due to constrained supply. In contrast, demand-pull inflation occurs when the economy’s productive capacity cannot meet aggregate demand. Such force typically emerges during expansionary periods, characterised by a sustained increase in demand, often attached to high levels of employment and output, and a shrinking output gap—the difference between actual and potential output.

Given that two key components drive inflation, it is vital to understand and distinguish between them (Hofmann et al. (2024)). Since these drivers differ in origin and persistence, their distinction is particularly relevant for targeted policy-making. Central banks’ responses to inflation depend on the nature of the inflation’s origin, making it crucial to accurately identify the sources of inflation (Clarida et al. (1999)). Moreover, central banks tend to respond more aggressively to demand-pull inflationary pressures by raising interest rates to control demand and contain prices, ultimately stabilising the business cycle (Hofmann et al. (2024)). In contrast, supply-side shocks require a more cautious approach, as such policies can harm the economy by slowing economic growth or inducing persistent inflation (Bańbura et al. (2023)). The need to balance stabilising inflation whilst preserving economic growth, requires making such difference when it comes to policy-making (Clarida et al. (1999); Hofmann et al. (2024)). Therefore, the source of inflation must be carefully evaluated and the policy response should be in accordance Bańbura et al. (2023)). This study investigates the extent to which improved VAR models can forecast inflation more accurately by distinguishing between cost-push and demand-pull shocks. Specifically, the central research question is: *How can advanced VAR methods better forecast and trace the impact of macroeconomic shocks on Eurozone inflation?* Particularly, the study examines the effects of supply-side and demand-side drivers. It aims to add value by distinguishing the effects of macro shocks on inflation and providing practical recommendations for policy responses that account for those distinctions.

In an increasingly interconnected world, capturing inflation’s causes and consequences has become more complex and critical. As economies are interrelated, domestic price dynamics respond more strongly to external shocks. This is particularly true in the Eurozone, where a unified monetary policy framework must navigate asymmetric shocks across member states (Arce et al. (2024)). Supply-side shocks—such as global supply-chain disruptions and energy-price surges triggered by the COVID-19 pandemic and the Russia–Ukraine war—have been major drivers of recent inflationary pressures (Ferrante et al. (2023); Shapiro (2022); Di Giovanni et al. (2022)). Consequently, production and transportation costs have risen sharply, with especially pronounced impacts in open and trade-dependent economies like the Eurozone (Bańbura et al. (2023)). Empirical studies con-

firm that these shocks impart persistence in inflation (Finck and Tillmann (2022); Eickmeier et al. (2023)). Against this backdrop, accurately distinguishing between the inherently different sources of inflation is more crucial than ever.

Consistent with this, this paper addresses this need by applying a suite of improved Vector Autoregressive (VAR) models to assess how macroeconomic shocks influence aggregate inflation over different time horizons. By doing so, this study will provide a framework that delivers more accurate inflation forecasts and deeper insights into the transmission mechanisms of cost-push and demand-pull shocks. Sims (1980) provided the groundwork for inflation modelling by introducing the VAR model. He argued against the overuse of restrictive identification in macroeconomic modelling and demonstrated that VARs might better reflect interdependence across variables than standard equation-by-equation models. However, traditional VARs encounter substantial issues when examining inflation dynamics: they do not distinguish structural shocks and become over-parameterised when several variables and lags are included, resulting in poor out-of-sample forecasting ability.

To address these limitations, the Structural VAR extends the standard VAR by imposing identification restrictions to distinguish cost-push from demand-pull shocks (Bańbura et al. (2023)). The Bayesian VAR (BVAR) framework addresses overfitting by introducing informative priors—pre-specified beliefs about model parameter values—to help regularise parameter estimates in high-dimensional settings. Combining both, Woźniak (2018) found that the Bayesian SVAR (BSVAR) model enhances structural shock recognition, estimation efficiency, and robustness in small sample sizes. Building on these advances, authors such as Giannone et al. (2015) and Litterman (1986) demonstrate how Bayesian techniques improve macroeconomic forecasting by incorporating prior assumptions and managing model uncertainty.

The research includes monthly data from January 2015 to January 2024, covering 14 macroeconomic variables controlling for supply-side, demand-side, and external factors influencing inflation in the Eurozone. The headline inflation rate is used as the inflation indicator, represented by the monthly growth rate of the Harmonized Index of Consumer Prices (HICP), which provides a consistent and comparable measure of overall price level changes across the Eurozone. The paper constructs and evaluates three scenarios for the SVAR and BSVAR frameworks to assess model performance under different structural assumptions. These scenarios include different variables designed to capture, explain, and quantify the effects of supply-side and/or demand-side shocks. The selection of variables for each scenario is based on theoretical relevance, statistical properties such as stationarity, and the absence of multicollinearity.

Building on this data framework, this paper extends the SVAR methodology. Specifically, it proposes a BSVAR model with Minnesota priors, motivated by Chan (2021), to improve the identification of cost-push and demand-pull shocks. Additionally, factor-augmented methods are employed to address high-dimensional macroeconomic data and improve shock transmission analysis (Agripino and Ricco (2018)). By incorporating robust probabilistic forecasting techniques, the paper aims to provide policymakers with a clearer understanding of inflation persistence under different scenarios.

While past research has focused on general inflation forecasting, this work explicitly decomposes aggregate inflation by capturing how different shocks propagate and persist over time. By integrating Bayesian estimation techniques, the study proposes a more reliable forecasting framework that provides early warning signals for persistent inflationary pressures. These approaches provide key insights as to whether enhanced VAR models better forecast the effect of various shocks on inflation and the persistence of these effects in the Eurozone. Specifically, the study investigates the extent to which BSVAR and factor-augmented approaches improve inflation forecasting relative to traditional SVAR models. Additionally, the impact of external shocks, such as global supply disruptions and commodity price surges, is examined. Thus, this paper contributes to the existing literature by improving the forecasting inflation models and understanding the dynamics of its distinct components. We aim to provide findings for more effective inflation targeting and policy-making, ultimately contributing to economic stability in the Eurozone.

The remainder of the paper is structured as follows: Section 2 provides a detailed description of the data used in the analysis. Next, Section 3 outlines the methodological framework employed to identify shocks and forecast inflation. Furthermore, Section 4 presents the empirical findings and model performance results. Lastly, Section 5 concludes the paper and discusses key insights for policy implications, the paper’s limitations, and future research.

2 Data

This paper includes several macroeconomic variables from January 2015 to January 2024 to effectively capture cost-push and demand-pull dynamics and explain the trends driving inflation. This approach distinguishes between temporary and persistent effects of the various shocks. Further, these variables will be used to study impulse responses and construct and evaluate forecasts. A detailed description of all these variables and their characteristics can be found in Appendix A.

Baseline Variables

Inflation tracks price-level changes in the Eurozone, guides monetary policy, and helps understand the lasting impact of economic shocks. In the resources sector, the paper includes Brent Crude Oil and Gas Prices. Crude and gas prices significantly impact production costs and consumer prices, as they have production and fuel purposes, directly increasing the cost of products. Baumeister and Peersman (2013) highlights the importance of oil prices in explaining inflation. Furthermore, the authors in Boeck and Zörner (2025) find that gas price shocks have significant effects on both inflation and its expectations. Moreover, studies show that gas is closely related to Brent, reinforcing the persistent nature of a resource shock (Kilian and Murphy (2014)). The EUR/USD exchange rate directly influences the cost of imported goods and services, with depreciations of the domestic currency making imports more expensive. A study by Goldberg and Knetter (1997) highlights that exchange rate fluctuations can have real-world consequences for trade dynamics and inflation. Disposable income controls for demand pressures on inflation; an increase in disposable income

is associated with increased consumer expenditure, stimulating economic activity and potentially contributing to inflation. By tracking disposable income, the paper can evaluate households' adjustments in spending in response to economic shocks. Notably, Dynan et al. (2004) provides evidence that changes in disposable income significantly affect consumer behaviour and broader economic dynamics, including inflation.

Non-Baseline Variables

Beyond the baseline variables that are generally included in inflation forecasting, the study extends with non-baseline variables to capture demand and supply specific patterns. This paper incorporates the GECON index to capture changes in global economic activity, thereby capturing exogenous conditions that influence Eurozone inflation dynamics. Moreover, the IP provides a benchmark the economic activity in the Eurozone, which helps to explain movements in inflation driven by changes in domestic activity. In addition, the GDP is included to control for the overall performance of the economy beyond the industrial sector. GDP includes more demand conditions, helping to explain a greater portion of shifts in inflation. Additionally, the LCI is captured to measure total labour cost per unit of output. Indeed, rising labor costs are often being passed on to consumers, hence contributing to inflation. Closely related, the unemployment rate helps capture the inverse relationship between inflation and labor market conditions, as suggested by the Philips Curve. To account for the influence of global supply dynamics, the paper includes the GSCPI index. It measures disruptions in global production and trade networks, which directly impact the cost of goods and services and therefore inflation. Moreover, interest rates are included to capture the effects of monetary policy, which is a key driver of inflation, on inflation dynamics. Furthermore, rising electricity prices directly affect production costs for companies, and they will often pass this burden on to consumers through higher prices for goods and services. Lastly, the Caldara-Iacoviello index captures geopolitical risks such as conflicts and political instability which in turn cause external inflationary shocks.

2.1 Data Preprocessing

Missing values

Given that the LCI and GDP series are observed only quarterly, the paper adopts a mixed-frequency state-space model and uses the Kalman filter–smoother to infer latent monthly values. The Kalman algorithm seen in Kalman (1960) yields minimum-variance estimates for unobserved states and has been shown to produce good monthly macro indicators or country-specific GDP. The authors in Guay et al. (1990) use Kalman filtering techniques to infer monthly values of the Canadian GDP for 1962-1985. This provides a sound basis for the interpolation in the scope of the study's research models. The estimation procedure embeds quarterly observations into a monthly framework by assigning the data every third month, estimating the process, and measuring noise variances via likelihood maximization. The Kalman filter is applied in a recursive manner: it first predicts the

latent state based on previous estimates, then updates these predictions when quarterly observations are available, weighing each according to its uncertainty. A subsequent smoothing step refines the filtered estimates by incorporating future information, thereby producing a robust interpolation of monthly values. The estimated monthly series are then combined with the rest of the monthly data framework.

Non-stationarity

A key assumption of the models used in this paper is stationarity of the data, with no regime switching from 2015 to 2024. Lucas (1976) highlights that non-stationarity can undermine the model’s reliability, as policymakers may alter structural relationships in response to significant data fluctuations. This is particularly important when using inflation directly in the model, as doing so may hinder the model’s ability to accurately capture long-term relationships. To address this issue, the paper conducts the Augmented Dickey-Fuller (ADF) test, confirming that most variables are non-stationary. Consequently, the variables are transformed using log-differences, first differences, or growth rates (see Appendix A). Since inflation does not fit the log-difference approach due to sign inconsistencies, the study focuses on its growth rate. The remaining variables that are still non-stationary after the transformation are excluded from the model.

Numerical instability

To address numerical instability caused by differences in variable scale, a standardization transformation is applied. This allows for a more accurate assessment of the relative impact of each shock on the impulse response function of inflation. While this approach is not yet standard practice in the literature, it becomes particularly important when variables differ in units of measurement, scale, or magnitude. As highlighted by Tena and Giovannoni (2015), standardizing or linearly transforming variables in VAR models can significantly improve the interpretability of impulse responses and the numerical stability of the model.

3 Methodology

This section starts by categorizing the variables from the dataset provided in Section 2 into demand-side and supply-side indicators, with two additional variables treated as external sources, as shown in Table 1. Following this, the paper introduces three approaches to forecast and analyze inflation dynamics driven by cost-push and demand-pull shocks: Structural Vector Autoregressions (SVAR), Bayesian SVAR (BSVAR), and a Dynamic Factor-Augmented VAR (DFAVAR) model. Additionally, the analysis considers three distinct scenarios: a baseline model incorporating both supply and demand variables, a model including only demand variables, and an extended model that adds two external factors beyond the first scenario. The methodological framework focuses exclusively on the first scenario, given the structural similarities among them. A detailed variable selection process is explained in Section 4.

Table 1: Supply-Side and Demand-Side Variables in the VAR Models

Supply-Side Variables	Demand-Side Variables
Brent Crude Oil Prices (π_{Brent})	Euro area inflation rate ($\%\Delta\pi$)
Gas Prices (π_{gas})	Exchange Rate EUR/USD ($\%\Delta E$)
Industrial Production ($\%\Delta IP$)	Unemployment rate (Δu)
Labour Cost Index ($\%\Delta LCI$)	Disposable income ($\%\Delta Y_{dis}$)
Global Supply Chain Pressure Index ($\Delta GSCPI$)	Caldara-Iacoviello index (Γ_{CII})
Electricity prices ($\%\Delta P_{elec}$)	Gross Domestic Product ($\%\Delta GDP$)

The model estimation relies on a carefully chosen subset of variables organized into three scenarios that balance relevance and data quality. In the base scenario, we include both the selected supply- and demand-side variables. The demand-only scenario restricts the model to demand-side factors. The extended scenario builds on the full specification by adding key external variables identified during the selection process. We pre-test all series for stationarity and eliminate multicollinearity to ensure the robustness of the estimates.

With the variables selected, the estimation follows a classical, frequentist strategy. As outlined by Kilian and Lütkepohl (2017), we implement the core VAR parameters via OLS, but face the familiar underidentification challenge when imposing long-run restrictions (see Chapter 11). Moreover, as the authors note, extending the system beyond three variables complicates VECM implementation. To address these issues in the richer dataset, we therefore complement the standard rank-trace test with economic priors when assessing stationarity and cointegration. A more detailed analysis of the cointegration relationships and stationarity properties of the variables will follow in subsequent sections, providing insights into their underlying dynamics.

3.1 Standard VAR Model Specification

The Standard VAR model specifies the economic variables at time t as a function of the lag variables in addition to the contemporaneous shock (Sims (1980)):

$$Y_t = A_1 Y_{t-1} + \dots + A_p Y_{t-p} + u_t, \text{ with } u_t \sim N(0, \Omega_{u_t}), \quad (1)$$

where Y_t is a $k \times 1$ vector of endogenous variables, A_i are $n \times n$ coefficient matrices, and u_t is a $k \times 1$ vector of white noise error terms with mean zero and covariance matrix Σ_u . Following this definition, the paper divides the Y_t vector into supply and demand, where Y_t^s represents an $m \times 1$ vector incorporating the supply-side variables, and Y_t^d represents a $(k - m) \times 1$ vector including the demand-side variables.

Nonetheless, the error terms u_t may exhibit contemporaneous correlations, indicating that the shocks are not inherently structural—that is, not directly interpretable in economic terms. To identify structural shocks and to analyze the contemporaneous effects of other endogenous variables

on inflation, the research extends the analysis to include additional model specifications.

3.2 Structural VAR Model

The Structural VAR model, proposed by Lütkepohl and Netšunajev (2017), extends the standard VAR model by imposing a structure on the error terms to identify structural shocks, denoted by ϵ_t . Specifically, the reduced-form errors u_t are related to the structural shocks ϵ_t through a structural impact matrix D such that $u_t = D\epsilon_t$, leading to uncorrelated error terms capturing true economic interpretation of shocks. Subsequently, after pre-multiplying the standard VAR specification by the D^{-1} structural matrix, the specification of the Structural Vector Autoregressive (SVAR) model is retrieved:

$$\begin{aligned} D^{-1}Y_t &= D^{-1}A_1Y_{t-1} + \dots + D^{-1}A_pY_{t-p} + D^{-1}u_t \\ &= B_1Y_{t-1} + \dots + B_pY_{t-p} + \epsilon_t, \end{aligned} \quad \text{where } B_i = D^{-1}A_i \text{ and } u_t = D\epsilon_t. \quad (2)$$

Simultaneity is a fundamental characteristic of macroeconomic models, capturing the inherent interdependence between variables. This feature is equally relevant in VAR models. While the standard VAR framework provides a relatively accurate representation of the dynamic lagged relationships between variables, it is subject to identification issues. Specifically, the error terms u_t exhibit correlation, complicating their interpretation within an economic context. Consequently, simultaneity results in shock effects absorbed by the error terms u_t rather than directly reflected in the endogenous variables. Although this formulation allows for the incorporation of indirect effects, it can conceal direct contemporaneous relationships among the endogenous variables. To establish a clear structure in the error terms, correctly identifying the structural matrix D is essential.

3.2.1 Identification of the Structural Matrix D

The reduced form errors u_t are linear combinations of structural shocks and therefore lack a direct economic interpretation. To isolate these underlying shocks, the paper identifies the structural matrix D , which links u_t to orthogonal structural shocks ϵ_t with $E(\epsilon_t\epsilon_t') = I_k$. The reduced-form error variance-covariance matrix Ω_u , estimated via OLS, has $k(k+1)/2$ unique elements, whereas the structural matrix D contains k^2 parameters. Therefore, $k(k-1)/2$ additional restrictions are needed for identification. These are typically imposed using short- and long-run restrictions.

Short-run zero restrictions

Motivated by Sims (1980), short-run identification is implemented using Cholesky decomposition on the reduced-form covariance matrix Ω_u . This method leverages the so-called "world ordering" approach, where variables in the VAR model are ordered based on their response speed to different shocks. The relationship between the reduced-form covariance matrix and the structural matrix is

given by:

$$\Omega_u = E[u_t u_t^\top] = D D^\top. \quad (3)$$

Applying Cholesky decomposition to Ω_u yields a lower triangular matrix L such that $\Omega_u = L L^\top$. By setting $D = L$, the resulting structural matrix adopts a lower triangular form. This recursive assumption imposes a natural ordering on the responses of variables to structural shocks. Specifically, the first variable responds only to its own shock, and the second variable responds to both its own shock and the first, but not later ones. This structure ensures that some shocks do not immediately affect certain variables, aligning with economic intuition. The ordering of variables in the model is instrumental in describing the causal relationships among them, governing which variables exert contemporaneous effects and which have lagged impacts. The paper’s approach prioritizes supply-side factors first, as their impact is immediate. For example, increases in oil or gas prices result in an instantaneous response in inflation, as highlighted by Asab (2025). Conversely, demand-side variables—specifically foreign disposable income and exchange rates—are positioned later. These external factors do not operate in the short run. Rather, their effects materialize only after a delay once the channels transmitting the shocks—like trade and exports—have fully absorbed the initial impacts (Obstfeld and Rogoff (2000)). After the structural matrix D is identified, the structural shocks can be recovered

$$\text{from } u_t = D \epsilon_t, \text{ it follows that: } \epsilon_t = D^{-1} u_t$$

Short-run sign restrictions

While zero-restrictions defined using Cholesky decomposition are effective for identification in the short run, they do not account for the dynamic and continuous interaction between variables, as dictated by economic theory. Specifically, there may be a consistently positive or negative short-run causal effect. To capture this, the identification of the structural matrix D in the SVAR model is achieved by imposing sign restrictions on the impulse responses, following the iterative guess-and-check procedure introduced by Kilian and Murphy (2012).

Indeed, some economic shocks are expected to cause only increases or decreases in one or more of the endogenous variables. For instance, an oil supply shock is expected to drive core inflation upwards. These expected directional responses form the basis for the sign restrictions, as outlined in Table 2 below, which shows the expected signs for shocks affecting the endogenous variables in Y_t . Based on the defined short-run restrictions, the contemporaneous effects among the variables are identified as follows, imposed as penalties when estimating the D matrix.

Table 2: Impact of Various Shocks on Different Variables

Variable / Shocks	Oil	Gas	Inflationary	Expansionary US income	Euro Appreciation
π_{Brent}	+	0	0	0	0
π_{gas}		+	0	0	0
$\% \Delta \pi$	+	+	+	0	0
$\% \Delta Y_{dis}$			+	+	0
$\% \Delta E$					+

Note: An entry with +/- denotes a positive/negative contemporaneous response of the variable to the specific shock. A 0 indicates no contemporaneous response and an empty cell denotes an unrestricted response.

Long-run restrictions

Building on the work of Blanchard and Quah (1988), who first applied long-run restrictions to output and the unemployment rate using a VAR framework, this analysis proposes extending this approach to analyze cost-push and demand-pull shocks. Specifically, it imposes long-run restrictions on all variables from both the supply and demand sides within the benchmark model to capture their persistence. This extension is motivated by the idea that demand-side shocks typically have a short-lived impact, whereas supply-side shocks have a more persistent and widespread effect on inflation. In the original paper, the focus is on the output gap and unemployment. In contrast, this paper incorporates all variables in the first scenario after addressing issues of cointegration and stationarity.

Starting from the SVAR process of order p defined in Equation 1, the study uses the lag operator L , defined as $L^j Y_t = Y_{t-j}$, and lag polynomial $A(L)$. The equation is rewritten compactly as

$$A(L)Y_t = u_t. \quad (4)$$

If the process is stable, meaning that the determinant $\det(A(L)) = 0$ has all its roots outside the unit circle, then $A(L)$ is invertible and can be expressed as an infinite power series:

$$A(L)^{-1} = \sum_{j=0}^{\infty} C_j L^j. \quad (5)$$

Consequently, multiplying both sides of equation 4 by $A(L)^{-1}$, we can obtain the moving Average (MA) representation is obtained.

$$Y_t = \sum_{j=0}^{\infty} C_j u_{t-j}. \quad (6)$$

This form expresses Y_t as an infinite sum of past shocks (see Appendix B), where the matrices C_j capture the dynamic response of the system to innovations. This MA representation is useful for studying impulse responses and forecasting in time series analysis.

The shocks captured in this reduced-form VAR can be divided into shocks on the supply and demand side. Identification relies on the assumption that only supply shocks have a lasting impact on supply-side factors, such as gas and oil prices. In contrast, demand shocks—originating from foreign disposable income and exchange rate fluctuations—are assumed to exert only transitory effects on these supply-side variables.

The identification strategy is summarized in a structured matrix form, where zeros are imposed to restrict the long-run influence of demand shocks on supply-side variables. The complete mathematical derivation of the model setup, including the formulation of the lag polynomial, its inversion, the MA representation, and the identification scheme, is provided in Appendix H.

3.2.2 Identification via Penalized Frobenius Factorization under Recursive Structure

To recover orthogonal structural shocks from the reduced-form residuals of a VAR, the paper considers a factorization of the form $\Sigma = LL^\top$, where Σ is the estimated residual covariance matrix and L denotes the contemporaneous impact matrix. It imposes a recursive structure by restricting L to the space $\mathcal{L}$ of lower triangular matrices, consistent with a Cholesky-type identification.

Rather than enforcing $\Sigma = LL^\top$ exactly, L is estimated by solving the penalized minimization problem:

$$\min_{L \in \mathcal{L}} \|\Sigma - LL^\top\|_F^2 + \lambda_1 \mathcal{P}_{\text{sign}}(L) + \lambda_2 \mathcal{P}_{\text{long}}(L),$$

where $\|\cdot\|_F$ denotes the Frobenius norm, and $\mathcal{P}_{\text{sign}}(\cdot)$ and $\mathcal{P}_{\text{long}}(\cdot)$ are differentiable penalty functions encoding, respectively, sign and long-run restrictions derived from economic theory. The scalars $\lambda_1, \lambda_2 \geq 0$ govern the strength of these soft constraints.

This formulation generalizes recursive identification by permitting deviations from exact equality $\Sigma = LL^\top$, thereby accommodating sampling variation and potential misspecification. The inclusion of penalty terms allows the incorporation of prior theoretical knowledge without imposing hard constraints, enhancing both the empirical robustness and interpretability of the structural decomposition. The resulting L preserves orthogonality and recursive ordering, while offering a statistically coherent approximation to the data-generating covariance structure.

3.3 Bayesian Structural Vector Autoregressions

The Structural VAR (SVAR) model serves as a fundamental framework for capturing the dynamic relationships between macroeconomic variables, as developed in papers Wang and McPhail (2014) and Dungey and Pagan (2000). However, SVAR models estimated using frequentist methods such as Maximum Likelihood Estimation (MLE) face several limitations, particularly in high-dimensional settings where overfitting and estimation uncertainty become prominent issues. Additionally, frequentist methods treat parameters as fixed but unknown and rely on large sampling variability and asymptotic properties for inference.

Unlike frequentist approaches, Bayesian methods as developed in Giannone et al. (2015) and Chan (2021) treat parameters as random variables with probability distributions, enabling quan-

tification of parameter uncertainty. By introducing a Bayesian Structural VAR (BSVAR) model, the priors are imposed for addressing the limitations of SVAR, particularly in handling structural breaks, preventing overfitting, and enabling probabilistic forecasts. This paper proposes two distinct Bayesian approaches -recursive and non-recursive- for structuring the SVAR model, each offering different advantages in terms of flexibility and identification.

The first approach is Non-Recursive BSVAR, which identifies structural shocks using various restrictions defined earlier for the SVAR framework. Unlike the recursive specification in the SVAR, this approach relaxes the assumption of a predetermined variable ordering, which allows for greater flexibility in model identification. This is particularly valuable in cases where the underlying economic structure does not adhere to a strict recursive framework. The second approach is the Recursive BSVAR, which aligns with the frequentist approach in the specification of the structural matrix. To capture the recursive relationships between variables, the paper employs the LKJ distribution introduced by Lewandowski et al. (2009). The LKJ distribution is used to identify positive-definite covariance matrices, characterized by a shape parameter. A covariance matrix Σ is obtained via the decomposition:

$$\Sigma = DRD^T \quad (7)$$

where D is a lower triangular matrix and R is a diagonal matrix, often the identity matrix. Details on the derivation of the probability density function of the identity matrix R can be found in Appendix E. The previously discussed restrictions remain valid for both Bayesian approaches, being implemented as tensor constraints during the matrix generation process.

The BSVAR model follows the same representation as the SVAR model while incorporating Bayesian priors, defined in Equation 2. To estimate the system, this paper introduces three main components: (1) prior distributions for coefficients matrix B_i , the structural matrix D and covariance matrix Σ_ϵ , (2) posterior inference, and (3) MCMC sampling.

Minnesota priors on coefficients B_i

Following the studies conducted by Giannone et al. (2015) and Chan (2021), this paper imposes the most commonly used prior in Bayesian analysis, Minnesota prior, to regularize the coefficients, which is essential for improving estimation efficiency and preventing overfitting in high-dimensional settings. The Minnesota prior assumes that each coefficient follows a normal distribution:

$$B_i \sim \mathcal{N}(0, V_{i,kk}),$$

where $V_{i,kk}$ is a diagonal variance matrix, which determines the degree of shrinkage applied to the coefficients. The specification of the variance structure follows the approach outlined by Chan (2021), and further details on the derivation can be found in the Appendix D. This variance structure ensures that the coefficients for higher lags are shrunk more heavily, preventing overfitting of distant lags that may not have meaningful economic relationships. The introduction of Bayesian priors allows for a model that can adjust to changing relationships over time, especially when there are significant

shifts in the economic environment.

Priors for scaling matrix D^{-1} and variance-covariance matrix Σ_ϵ

In addition to priors for coefficients, the paper also introduces priors for D^{-1} and Σ_ϵ . The former represents a scaling of the endogenous variables, constrained by identification strategy, and the latter is the variance-covariance matrix of the structural errors.

A common choice for these priors is the Inverse-Wishart distribution (Agrippino and Ricco (2018)). This distribution is commonly used for covariance matrices in Bayesian models because it is a conjugate prior. As the analysis considers a SVAR model, the priors can be written as:

$$D^{-1} \sim IW(\Psi_0, \nu_0) \text{ and } \Sigma_\epsilon \sim IW(\Psi_0, \nu_0).$$

Here, Ψ_0 represents the scale matrix, which corresponds to the prior beliefs about the scale and structure of the covariance matrix, while ν_0 denotes the degrees of freedom parameter that determines how informative the prior is. For instance, a large ν_0 gives a more diffuse prior, while a small ν_0 implies stronger belief in the scale matrix Ψ_0 . The choice of Ψ_0 and ν_0 depends on prior beliefs about the structure of D^{-1} and Σ_ϵ . When little prior knowledge is available, the weakly informative prior is chosen to ensure that the model remains flexible and that the data primarily drives the estimation process. Particularly, the paper assumes each variable has a unit variance and that there is no prior preference for any particular correlation structure by setting Ψ_0 equal to the identity matrix. Similarly, setting $\nu_0 = k + 1$ ensures that the prior is minimally informative while still being a valid Inverse-Wishart distribution, with k being the dimension of the matrix.

Identification in SVAR models revolves around the estimation of D^{-1} through external economic restrictions. In the Bayesian approach, rather than estimating the coefficients directly, this paper imposes prior probability distributions.

Posterior distribution

In a Bayesian context, the posterior distribution of the parameters in the SVAR model can be derived using Bayes' theorem, which combines the likelihood of the data with the prior beliefs about the coefficients, error covariance, and scaling matrix. The parameters needed to be estimated are coefficient matrices B_i , scaling matrix D^{-1} and covariance matrix of structural shocks Σ_ϵ . The paper assumes that the errors ϵ_t are normally distributed with mean 0 and covariance matrix Σ_ϵ . The likelihood function expression can be found in Appendix F. Based on the Bayes' Theorem, the posterior distribution of the parameters given the data is proportional to the product of probability of observing the data given the parameters and the prior beliefs about B_i, D^{-1} and Σ_ϵ (Giannone et al. (2015)). In a Bayesian analysis, the parameters are estimated by drawing samples from this posterior distribution using Markov Chain Monte Carlo methods.

Markov Chain Monte Carlo (MCMC) sampling

Since the posterior distribution cannot generally be computed analytically due to the complex structure of the model, the paper uses Markov Chain Monte Carlo (MCMC) methods, specifically Gibbs sampling, to generate samples from the posterior distribution. The central idea of MCMC process is to draw a sequence of samples from the joint posterior distribution of the parameters $\theta = (B_1, B_2, \dots, B_p, D^{-1}, \Sigma_\epsilon)$, generating a Markov chain that eventually converges to the target posterior distribution. Motivated by Agrippino and Ricco (2018), Gibbs sampling involves iteratively sampling from the conditional distributions of each parameter, given the current values of the others. More specifically, the paper samples from the conditional distribution for B_i given D^{-1} and Σ_ϵ , for D^{-1} given B_i and Σ_ϵ , and for Σ_ϵ given B_i and D^{-1} .

The conditional distribution of the coefficient matrix B_i given the inverse of the structural matrix D^{-1} and the variance-covariance matrix Σ_ϵ follows a normal distribution, shown as follows:

$$B_i \mid Y, D^{-1}, \Sigma_\epsilon \sim \mathcal{N}(M_B, V_B), \quad (8)$$

where

$$V_B = (X^\top X + \Sigma_B^{-1})^{-1}, \quad M_B = V_B X^\top Y. \quad (9)$$

Meanwhile, the conditional distributions of D^{-1} and Σ_ϵ both follow an Inverse-Wishart distribution, whose parameters have been updated by the data and the prior.

After a sufficient number of iterations, the MCMC process converges to the true posterior distribution with a vector of N posterior samples $\theta^{(1)}, \theta^{(2)}, \dots, \theta^{(N)}$, which could be used to estimate the posterior means of the parameters and compute confidence intervals from the posterior samples:

$$\hat{\theta} = \frac{1}{N} \sum_{k=1}^N \theta^{(k)}, \quad \text{and} \quad CI_\theta = (\theta_{(2.5\%)}, \theta_{(97.5\%)}). \quad (10)$$

While the BSVAR model improves upon the SVAR model by incorporating Bayesian priors to prevent overfitting, it still faces challenges related to multicollinearity when a large number of variables is considered in the model. To address this limitation, the paper extends the BSVAR framework to a Dynamic Factor Augmented VAR (DFAVAR) Model in the following subsection.

3.4 Dynamic Factor Augmented Vector Autoregressions

In practice, all the previously mentioned SVAR methods break down when the assumption that the structural matrix D is invertible fails. This reliance can result in misleading outcomes that fail to accurately capture the true behavior of inflation. The problem becomes especially pronounced when dealing with a large number of endogenous variables, which often introduces multicollinearity.

Subsequently, time series models frequently suffer from the "curse of dimensionality" in such cases, leading to overparameterization and estimation inefficiencies. To address this, prior research

has proposed the use of factor-augmented models, which condense the common variation across many time series into a smaller set of latent factors. This approach reduces the model's dimensionality while retaining essential macroeconomic dynamics. A prominent example is the Factor-Augmented VAR (FAVAR) model developed by Bernanke et al. (2005).

Dynamic Factor Model Specification

The dynamic factor model assumes that the observed variables, denoted as Y_t , can be decomposed into a small number of dynamic factors f_t and an idiosyncratic component v_{it} . Specifically, let Y_t be an $k \times 1$ stationary vector observed over $t = 1, \dots, T$. Assume both k and T are large. The latent factors f_t , a $q \times 1$ vector, capture the common sources of variation in the observed variables. This research sets $q = 2$ to distinguish between the two types of shocks, assigning supply-side variables to one factor and demand-side variables to the other.

The model assumes that each component $y_{i,t}$ of Y_t follows the factor structure

$$y_{i,t} = \tilde{\lambda}_i(L)f_t + v_{i,t}, \quad (11)$$

where $\tilde{\lambda}_i(L)$ is a lag polynomial of order p . The idiosyncratic component $v_{i,t}$ may exhibit serial correlation, modeled as

$$v_{i,t} = \delta_i(L)u_{i,t-1} + w_{i,t}, \quad (12)$$

where $\delta_i(L)$ is a lag polynomial (i.e. a finite-order lag operator) and $w_{i,t}$ is a white noise process.

The dynamic factors f_t follow a VAR process, modeled as

$$f_t = \Gamma(L)f_{t-1} + \eta_t, \quad (13)$$

where $\Gamma(L)$ is a lag polynomial, and η_t represents a vector of factor innovations.

In this factor model, all of the cross-series co-movement is attributed to the common factors: the idiosyncratic shocks have zero contemporaneous correlation with one another, and they are also uncorrelated over time. Moreover, the factors themselves are assumed to be orthogonal to these idiosyncratic components so that no leftover correlation remains outside the factor structure. This setup is known as an exact dynamic factor model, in which all covariance among variables is fully explained by the factors. It is exact because of the first assumption, and is dynamic because both current and lagged factors affect y_{it} .

To facilitate estimation and forecasting comparison with other models, this paper expresses the dynamic factor model in a VAR form. This is achieved by substituting the factor process into the measurement equation and then stacking the system to include both observable variables and latent factors. The resulting specification is a joint VAR in the observed variables and lagged factors, shown in Equation 14:

$$\begin{bmatrix} F_t \\ Y_t \end{bmatrix} = \begin{bmatrix} \Phi(L) & 0 \\ \Lambda\Phi(L) & D(L) \end{bmatrix} \begin{bmatrix} F_{t-1} \\ Y_{t-1} \end{bmatrix} + \begin{bmatrix} I & 0 \\ \Lambda & I \end{bmatrix} \varepsilon_t, \quad \varepsilon_t = G\eta_t + w_t. \quad (14)$$

The full mathematical derivation of this transformation into a VAR representation, including the construction of the block-diagonal matrix and augmented factor vector, is presented in Appendix N.

3.5 Impulse Response Analysis of Inflation Dynamics and Forecast Evaluation

Once the structural shocks are identified using all the approaches mentioned in the previous subsections, the paper examines their impact on the rate of inflation using Impulse Response Functions (IRFs), defined as follows:

$$\text{IRF}_{\pi,s}(h) = \frac{\partial \pi_{t+h}}{\partial s_t}, \quad (15)$$

where π_{t+h} represents the inflation after h periods of shocks. Using the IRFs allows the research to analyze how inflation reacts to different economic shocks in the short term and identify which shocks have the most persistent effects. Additionally, varying the forecast horizon helps determine whether inflation adjusts gradually over time or experiences an immediate response to a given shock.

Furthermore, this research evaluates model performance using evaluated using Root Mean Squared Error (RMSE) that quantifies the average magnitude of forecast errors. To evaluate and compare model performance, the RMSE is computed for one-step-ahead, three-step-ahead, and six-step-ahead forecasts across all modeling approaches. Specifically, the study first compares forecast accuracy within the SVAR and BSVAR frameworks across the three defined scenarios. Following that, it compares the performance of the BSVAR and DFAVAR models.

4 Results

As described in the previous section, numerical experiments on the models were conducted using Python 3.12.4 on an Apple M1 2020 with 8 GB of RAM. The objective is to explore how macroeconomic shocks, both from the demand side and supply side, affect inflation across different time horizons using improved VAR models: Structural Vector Autoregression (SVAR), Bayesian SVAR (BSVAR), and Dynamic Factor-Augmented VAR (DFAVAR). This section introduces the three model scenarios, where variables are selected based on their theoretical relevance, stationarity, and lack of multicollinearity. For the SVAR model, the study first examines how shocks to one variable influence others within the system, considering both short- and long-term effects. Consequently, the Impulse Response Functions (IRFs) for inflation in response to various shocks are presented across all three scenarios. Following this, the study introduces the point estimates and distributions of the parameters obtained using the Bayesian approach and compare the corresponding IRFs to the SVAR model. Additionally, a fourth scenario is introduced that incorporates more supply-side variables to enable a comparison between the BSVAR and DFAVAR models. Finally, the study evaluates the out-of-sample forecasting accuracy of each model using the Root Mean Squared Error (RMSE).

Based on this analysis, the analysis aims to determine which model provides the best predictions of inflation and offers insights into how demand-pull and cost-push factors drive inflation dynamics.

4.1 Variables Selection

The selection of variables for the three scenarios is based on their theoretical relevance, stationarity properties, and lack of multicollinearity. Each variable is chosen for its economic significance in explaining inflation dynamics. The research conducts the Augmented Dickey-Fuller (ADF) test for stationarity over the estimation period from January 2015 to December 2022 on all variables in the dataset. Some were found to be non-stationary, even after transformations. As stationarity is a critical assumption for VAR-based models, these non-stationary variables were excluded from the analysis.

The correlation matrix of the remaining variables is shown in Appendix M, where the study selects those that are not highly correlated with each other to avoid multicollinearity and cointegration issues. In Scenario 1, variables from both the demand and supply sides are included. All the selected variables are stationary. Brent Crude Oil Prices and Gas Prices are included as key supply-side indicators as these are commonly identified as primary drivers of cost-push inflation. US Disposable Income and the FX rate are included to capture demand-side dynamics, reflecting changes in consumer spending and exchange rate movements.

Scenario 2 isolates the impact of demand-side shocks by excluding the supply-side variables present in Scenario 1, and Scenario 3 expands on Scenario 1 by incorporating the *GECON* index and interest rate to account for the role of global economic activity and central bank policy in shaping inflationary pressures. Table 3 summarizes the different scenarios.

Table 3: Scenario Composition

Scenario	Description	Variables Included
1	Supply and Demand	Brent Prices, Gas Prices, EU Inflation, US Disp. Income, FX Rate
2	Demand Only	EU Inflation, US Disposable Income, FX Rate
3	Supply, Demand and External Factors	Brent Prices, Gas Prices, EU Inflation, GECON Index, Interest Rate, US Disp. Income, FX Rate
4	All Selected Variables	Brent Prices, Gas Prices, EU Inflation, GECON Index, Interest Rate, US Disp. Income, FX Rate Industrial Production, Labour Cost Index, Interest-Rate Differential, Caldara-Iacoviello Index

During the stationarity tests over the estimation period, the study found that GDP, unemployment rate, and GSCPI were non-stationary, even after transformation. Consequently, these variables were excluded from the models. As seen in Figure 14 in Appendix M, several variables remain for potential inclusion, which will be considered when estimating the DFAVAR model.

4.2 Standard Vector Autoregressive Model

Despite the widespread use of standard VAR models due to their simplicity, the RMSE results found in Appendix G display unconventional trends. From 1-step to 6-step projections, the RMSEs

drop as the forecast horizon rises. This trend contradicts widely accepted macroeconomic forecasting principles, which indicate that error values usually rise with time because of accumulating uncertainty. The declining RMSE trend implies that the standard VAR might be overfitting short-term noise or missing the actual structural dynamics driving inflation, thus producing misleadingly improved forecasts at longer horizons.

The model behaves erratically in all scenarios. The RMSE is relatively high at the 1-step forecast in Scenarios 1 and 3, which both include supply-side and external variables. However, the RMSE decreases by the 6-step horizon. Scenario 2 (demand side alone), on the other hand, starts with an abnormally low 1-step RMSE and increases marginally with the horizon, although it is still below the others. Because it responds inconsistently to the input set’s structural composition, this behavior implies that the standard VAR is not robust to variable inclusion. This discrepancy highlights the VAR’s shortcomings in accurately modeling the multi-causal dynamics of inflation, particularly in the presence of both supply-side and demand-side variables.

Ultimately, these findings validate the claim that traditional VARs are inadequate for decomposing inflation into structurally relevant elements. They may fit short-term variations due to their statistical simplicity, but they cannot identify and model the economic mechanisms required for reliable and accurate inflation forecasting. Scenarios 1 and 3’s relatively poor performance in early timeframes further suggests that the VAR framework has trouble simulating more intricate inter-variable connections, where structural identification is crucial. These anomalies highlight the VAR model’s inability to model and capture inflationary dynamics properly. This reinforces the need for models incorporating structural identification and prior information to yield more reliable results with statistical robustness.

4.3 Structural Vector Autoregressive Model

To determine the optimal lag length for each scenario, the research estimates standard VAR models and compares the Akaike Information Criterion (AIC), Hannan-Quinn Criterion (HQ), and Bayesian Information Criterion (BIC), which balance model fit and parsimony. AIC suggests a lag of 7, HQ recommends 2, and BIC points to 1. Given the limited sample size and the need to avoid overfitting, the study adopts the BIC’s more conservative recommendation of a single lag, which is well-suited for maintaining robustness in SVAR and BSVAR analyses.

4.3.1 Structural Identification

To estimate the structural model, the study imposes a set of identification restrictions as outlined in Section 3. These restrictions help recover the structural matrix D from the reduced-form representation. This matrix captures the transformation between these two types of shocks and reflects the contemporaneous interactions among variables. By applying a recursive identification scheme in each scenario, the paper assumes that a structural shock can only contemporaneously influence the variables ordered after its corresponding variable in the system.

In the main text, the structural matrix and long-run impact matrix are presented for Scenario 1, as this scenario represents a balanced specification including both demand-side and supply-side variables. Scenarios 2 and 3, while still relevant, are either more restricted or extended. The detailed identification matrices for these scenarios can be found in Appendix I.

Table 4: Structural Matrix D capturing relations between structural shocks and innovations for Scenario 1

	π_{Brent}	π_{gas}	$\% \Delta \pi$	$\% \Delta Y_{dis}$	$\% \Delta E$
π_{Brent}	0.9175	0.0000	0.0000	0.0000	0.0000
π_{gas}	-0.0006	0.9517	0.0000	0.0000	0.0000
$\% \Delta \pi$	0.0426	-0.0001	0.9912	0.0000	0.0000
$\% \Delta Y_{d,US}$	-0.0600	-0.0889	-0.0358	0.7956	0.0000
$\% \Delta E$	0.0533	0.1093	0.2017	-0.2191	0.8761

Note: Each entry denotes the magnitude of a contemporaneous response of the (column) variable to the specific shock (row).

Table 4 presents the estimated structural matrix D for Scenario 1, which captures the contemporaneous relationships between structural shocks and reduced-form innovations. The identification strategy relies on sign restrictions and a recursive structure imposed via Cholesky decomposition, resulting in a lower triangular matrix. In this framework, the ordering of variables determines the assumed contemporaneous causal structure: structural shocks to variables listed earlier in the ordering can contemporaneously affect those that follow, but not vice versa. Since Brent Crude Oil Prices are placed first in the ordering, their shocks can contemporaneously influence all other variables in the system. In contrast, shocks to the EUR/USD exchange rate are contemporaneously influenced by all preceding shocks.

Notably, the diagonal elements of this matrix are close to 1, which indicates that each structural shock is largely explained by its own reduced-form shocks. Further, the coefficient of -0.0006 on the Brent oil price shock in the gas price equation suggests a very small negative contemporaneous effect. This finding is consistent with the low empirical correlation (0.07) observed between Brent oil and gas prices shown in the correlation matrix in Appendix M. Additionally, when considering the contemporaneous impact of both cost-push factors on CPI inflation, the study observes small positive effects: 0.0426 for Brent and -0.0001 for Gas Prices, which explains the theoretical expectation of cost-push inflation, where increases in input prices contribute to upward pressure on consumer prices.

4.3.2 Long-Run Impact Analysis

While the structural matrix shown in Table 4 captures the contemporaneous effects of structural shocks, Table 5, denoted as $C_j D$ in Section 3, reflects the cumulative effects of each structural shock on all variables in the system. These long-run impacts can be interpreted in two ways. First, the

diagonal elements being close to or greater than 1 indicate that a variable’s own shock has a persistent long-term effect. Second, while the off-diagonal elements represent spillover effects, capturing how a shock to one variable influences others in the system over time.

Given the focus on long-term inflation dynamics of macroeconomic shocks, special attention is given to the third row of the matrix, which captures long-run drivers. The results suggest that cost-push shocks, particularly those from Brent oil prices and gas prices, have moderate but persistent effects on inflation over the long run. Notably, while the Brent shock has a positive influence (0.1589), the effect of gas prices is slightly negative (-0.1435). Conversely, demand-side shocks—represented by disposable income and the exchange rate exhibit ignorable long-run influence on inflation, as their corresponding entries in the third row are close to zero.

Table 5: Long-Run Impact Matrix for Scenario 1

	π_{Brent}	π_{gas}	$\% \Delta \pi$	$\% \Delta Y_{dis}$	$\% \Delta E$
π_{Brent}	1.3577	-0.1100	-0.0705	0.0085	0.2348
π_{gas}	0.3514	1.0848	-0.0090	-0.0905	0.2401
$\% \Delta \pi$	0.1589	-0.1435	0.9239	0.0133	-0.0607
$\% \Delta Y_{dis}$	-0.2138	-0.1055	-0.0022	0.4948	0.0053
$\% \Delta E$	-0.0862	-0.1622	0.2970	-0.2728	1.1999

Figure 1 presents the impulse responses of the inflation growth rate to shocks originating from the variables in Scenario 1. The results confirm that supply-side shocks exert moderate but persistent effects, with their impacts remaining longer. In contrast, demand-side shocks cause a significant initial spike in inflation growth, but their effects drop quickly after around 70 periods. The speed of convergence to zero is notably faster for demand-side shocks, whereas supply-side shocks maintain a more sustained influence on inflation. Ultimately, the effects of all shocks fade away completely.

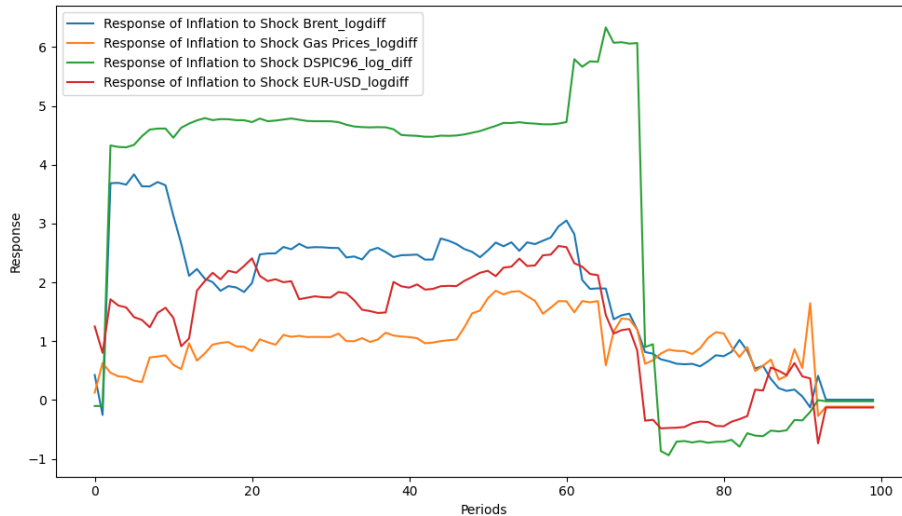


Figure 1: Impulse responses of Scenario 1

Similar patterns emerge in Figure 2 and 3, which visualize the impulse responses of inflation to shocks in Scenario 2 and Scenario 3. As in Scenario 1, supply-side shocks exhibit a persistent influence on inflation over time. However, demand-side shocks show a more temporary effect by causing an immediate spike in inflation that gradually fades.

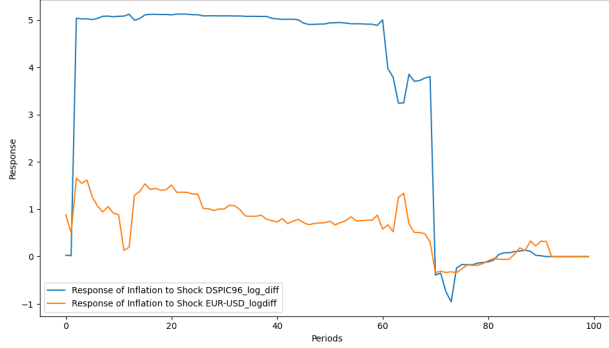


Figure 2: Impulse responses of Scenario 2

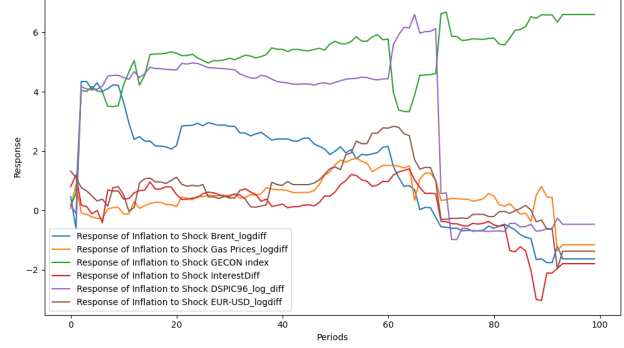


Figure 3: Impulse responses of Scenario 3

Specifically in Figure 3, *GECON* index shocks have a substantial and persistent effect of shocks on inflation. This index captures broader external factors such as global economic growth, international trade disruptions, and commodity price shifts. It is then observable that such shocks appear to have a dominant and long-lasting effect on inflation in the Eurozone. The impulse responses show that the effects of these shocks reach their maximum magnitude after approximately sixty months and remain significant even after 100 months, reinforcing the persistent nature of cost-push shocks on inflation. Notably, this effect surpasses any domestic supply or demand shocks, suggesting that global economic conditions play a crucial role in determining inflation dynamics.

Unlike the previous scenarios, it is also noticeable that not all shocks in Scenario 3 converge to zero, even after 100 months. Specifically, the interest rate shock, representing monetary policy actions, negatively affects inflation in the long run. This result suggests that tight monetary policies have a delayed but persistent adverse effect on inflation, with the influence lingering even after a long horizon. The adverse effect of interest rate shocks is consistent with the economic theory that higher interest rates reduce inflationary pressures over time by cooling consumption and investment. This pattern suggests that global economic shocks dominate in the short to medium term. However, monetary policy actions can have lasting effects on inflation in the long run, though less immediate. These findings underscore the role of domestic monetary policy measures and external factors in shaping inflation outcomes in the Eurozone.

4.3.3 SVAR-based Inflation Forecasts and Accuracy Evaluation

To evaluate the forecasting performance of the SVAR model, the study employs a one-step-ahead rolling window forecast of the inflation growth rate for each of the three scenarios. The window size is set to 95 to allow the model to be estimated on a fixed sample from January 2015 to December 2022, after which the window is shifted step by step to generate predictions for January 2023 to

December 2023. Figure 4 below compares each scenario's forecasted inflation growth rates with the actual observed values over this period. The study observes that all three scenarios generally track the evolution of actual inflation reasonably well and capture the directional patterns well. The forecasting pattern from all scenarios could be assessed in three parts.

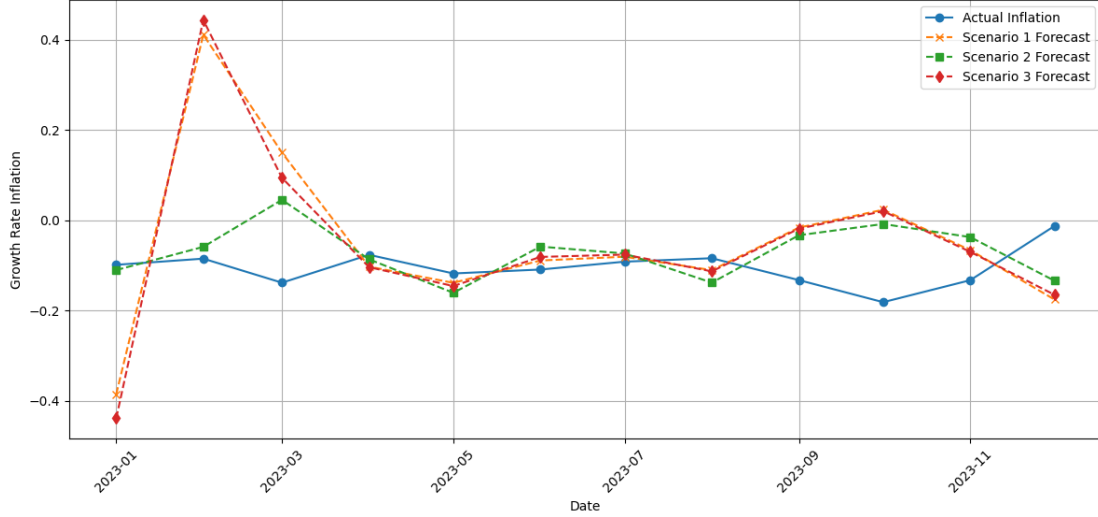


Figure 4: Growth rate inflation forecast of all scenarios vs actual growth rate inflation

First, at the beginning of 2023, all scenarios exhibit a notable upward spike. The spike is most pronounced in Scenarios 1 and 3, where the key supply-side variables are included. This spike could be attributed to the high volatility observed in late 2022, driven by remaining post-pandemic supply chain shocks that propagate into early 2023 due to the lag structure of the SVAR model. This pattern aligns with the plots of IRFs presented in Figure 1 and 3, demonstrating that these supply-side variables exert persistent and significant effects on inflation. Furthermore, during the middle phase of the year, forecasts across all scenarios closely align with actual inflation. This improved performance is likely due to the increasing stability of the post-pandemic economic environment, which results in more stable data and reduced influence of earlier outlier observations. Lastly, forecast performance weakens again in the year's final quarter as the forecasts fail to capture the observed downward shift in the actual growth rate inflation. This divergence could arise from unexpected shocks or structural changes not captured by the model.

Although the differences in forecast accuracy across the three scenarios are relatively modest, Scenario 2 consistently exhibits slightly better performance. This finding is reinforced by the Root Mean Squared Error (RMSE) values reported in Table 6, which assess forecast accuracy across different time horizons. These measures provide insight into the robustness of inflation forecasts over short- and medium-term periods. From the table, Scenario 2 yields the lowest RMSE at the one- and three-step-ahead horizons, highlighting its relative strength in capturing short-term inflation dynamics. This finding supports the earlier interpretation that demand-side variables may shape inflation dynamics in the short run. Interestingly, all three scenarios converge to similar RMSE values at the six-step-ahead horizon, and the relative advantage of Scenario 2 diminishes over longer

horizons. This convergence is consistent with the impulse response analysis discussed earlier, where supply-side and global shocks exhibit more persistent but delayed effects on inflation. This finding also supports the claim that demand-pull forces have a decreasing influence on inflation with time. Therefore, a more parsimonious SVAR model focusing on demand-side variables may yield more reliable forecasts in the short run. Meanwhile, the expanded specification in Scenario 3 does not improve forecast accuracy and may weaken predictive precision due to overfitting.

Table 6: Forecast RMSE for Scenario 1-3 using SVAR

Horizon	Scenario 1	Scenario 2	Scenario 3
1	0.2042	0.0940	0.2109
3	0.0799	0.0786	0.0826
6	0.0837	0.0842	0.0837

4.4 Bayesian Structural Autoregressive Model (BSVAR)

4.4.1 Identification of Structural Matrix: Recursive and Non-Recursive Approaches

As described in Section 3, the relationship between structural shocks in the BSVAR model is captured by imposing prior distributions on the coefficients matrix and then sampling from the posterior distribution using a Markov Chain Monte Carlo (MCMC) algorithm. This process generates a distribution of coefficient estimates. For clarity and interpretability, the mean of these estimates is taken only for Scenario 1. The results for the structural matrix D are presented in Table 7, representing the non-recursive and recursive Bayesian approaches, respectively.

Table 7: Point Estimates for Structural Matrix D for Recursive and Non-Recursive BSVAR in Scenario 1

	Non-Recursive BSVAR					Recursive BSVAR				
	π_{Brent}	π_{gas}	$\%\Delta\pi$	$\%\Delta Y_{dis}$	$\%\Delta E$	π_{Brent}	π_{gas}	$\%\Delta\pi$	$\%\Delta Y_{dis}$	$\%\Delta E$
π_{Brent}	-0.4030	0.1269	-0.4079	0.1246	-0.0081	1.0299	0.0000	0.0000	0.0000	0.0000
π_{gas}	0.1647	-0.3292	-0.3466	0.1580	0.0197	0.0228	1.0050	0.0000	0.0000	0.0000
$\%\Delta\pi$	0.5193	0.4989	0.1691	0.4767	0.3880	0.0727	0.0828	0.9690	0.0000	0.0000
$\%\Delta Y_{dis}$	0.1078	0.0920	-0.4714	-0.5296	-0.0524	0.0712	0.0994	0.0421	1.2063	0.0000
$\%\Delta E$	-0.2655	-0.2299	-0.1516	-0.4435	0.2739	-0.0319	-0.0937	-0.1985	0.2762	1.0905

From these tables, the study observes that, in the non-recursive approach, the structural matrix D is fully parameterized in order to allow for contemporaneous interaction among all variables. In contrast, the recursive specification imposes a lower triangular structure on D , which reflects a recursive identification strategy where each variable is contemporaneously affected only by those ordered before it. Analyzing the results, the growth rate of inflation is largely driven by its own shocks

(0.9690) in the recursive model. Further, the effect of cost-push factors appears to be relatively small, which aligns with earlier findings using the SVAR model in Table 4. However, in the non-recursive setup, inflation is significantly affected by Brent oil and gas prices (0.5193 and 0.4989) and also shows notable contemporaneous feedback from disposable income from the USA (0.4767) and the exchange rate (0.3880). This suggests that the non-recursive estimation captures a more complex and multi-directional relationship, which potentially offers a more flexible depiction of interdependencies among the variables.

While the SVAR framework requires a recursive ordering to identify structural shocks, the non-recursive BSVAR allows for more flexible identification without such restrictions. However, allowing for richer contemporaneous relationships goes at the cost of increased estimation complexity and the potential risk of overfitting. In the following subsections, the paper compares the impulse response functions derived from both BSVAR approaches across the three scenarios to assess whether recursive ordering improves the interpretability and robustness of the results.

4.4.2 Comparison of Impulse Responses Across Scenarios

Figures 5 and 6 compare the impulse responses of the growth rate of inflation to various structural shocks using the non-recursive and recursive BSVAR approaches under Scenario 1. A clear difference from the SVAR model emerges: in this BSVAR framework, both supply and demand shocks generate moderate but immediate responses in inflation, which decay rapidly within just a few periods. This is in contrast to the SVAR model discussed earlier, where the effects of such supply shocks often linger and diminish slowly, sometimes persisting up to 70 periods. This difference can be explained by the regularization properties of the Bayesian approach. In BSVAR, prior distributions are imposed on the parameters, which helps constrain implausible dynamics and reduces the persistence of shocks that may arise due to overfitting or multicollinearity in SVAR models. As a result, the BSVAR model produces sharper, more realistic responses that reflect the inherent uncertainty and variability in macroeconomic relationships.

Moreover, while demand-side shocks, such as those from disposable income and exchange rate fluctuations, have traditionally been considered to have strong, immediate effects on inflation, the BSVAR model reveals moderate short-term impacts. This may stem from the way Bayesian estimation combines observed data with economic theory, allowing even demand shocks to have less influence when supported by the data.

The non-recursive BSVAR shown in Figure 5, uses a fully parameterized structural matrix which enables contemporaneous interactions among all variables. This flexibility captures a complex, multidirectional system in which inflation is simultaneously influenced by shocks to commodity prices, income, and exchange rates. On the other hand, the recursive BSVAR shown in Figure 6 enforces a lower-triangular structure on the matrix of contemporaneous relationships. Here, inflation can only respond contemporaneously to variables that precede it in the assumed ordering. While this identification strategy is easier to interpret and grounded in traditional macroeconomic theory, it might underestimate the actual influence of contemporaneous shocks due to its structural rigidity.

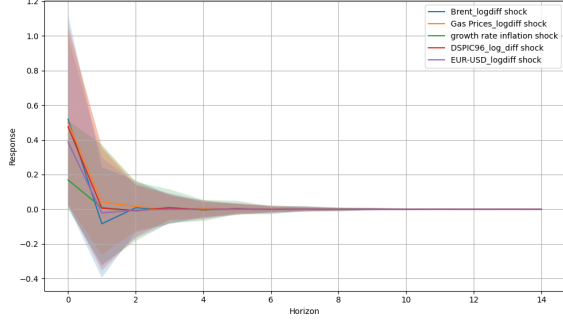


Figure 5: Impulse responses of Scenario 1 using non-recursive ordering

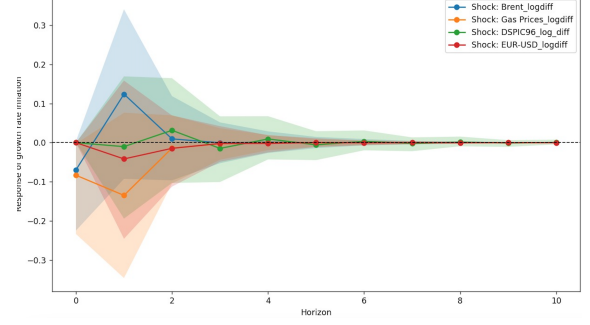


Figure 6: Impulse responses of Scenario 1 using recursive ordering

4.4.3 BSVAR-based Inflation Forecasts and Accuracy Evaluation

Table 8 presents the Root Mean Squared Error (RMSE) for forecast horizons of 1, 3, and 6 steps ahead, comparing the performance of recursive and non-recursive BSVAR models under three distinct model configurations. Although the recursive BSVAR generally exhibits inferior forecasting performance relative to the non-recursive specification, several nuanced findings emerge upon closer examination.

In the first scenario—characterized by a balanced inclusion of both demand and supply-side variables—the non-recursive model yields the lowest forecast errors across all horizons. This result suggests that a mixed-variable configuration may pose challenges for the imposition of a valid recursive causal structure, possibly due to the complexity in capturing dynamic interdependencies between demand and supply factors within a recursive identification scheme.

In contrast, Scenario 2 includes only demand-side variables. Under this specification, the recursive BSVAR outperforms the non-recursive model across all horizons. This finding implies that, when the system is restricted to a limited and homogeneous set of variables (i.e., three demand indicators), the recursive identification strategy more accurately captures the underlying causal structure, thereby improving forecast accuracy.

Scenario 3 incorporates five demand-side and two supply-side variables. While the recursive model underperforms at shorter horizons, its relative accuracy improves over time and surpasses the non-recursive model by the 6-step horizon. This indicates that the recursive identification scheme may better capture long-term structural relationships, particularly those associated with supply-side dynamics, whose effects tend to unfold over extended horizons.

Moreover, a cross-scenario comparison reveals that, at shorter horizons, the specification including only demand-side variables (Scenario 2) yields superior forecast performance. However, as the forecast horizon extends to six steps, models incorporating supply-side variables (Scenario 3) begin to outperform, suggesting a temporal divergence in the influence of macroeconomic drivers. Specifically, demand-side variables appear to dominate inflation dynamics in the short run, while supply-side variables exhibit a more persistent and accumulative influence over the long term. This observation aligns with economic theory that attributes short-term price fluctuations predominantly

to demand shocks, while recognizing the enduring role of supply-side factors in shaping inflationary trends.

Table 8: RMSE Values — Non-Recursive vs. Recursive BSVAR

Horizon	Non-Recursive			Recursive		
	Scenario 1	Scenario 2	Scenario 3	Scenario 1	Scenario 2	Scenario 3
1	0.1655	0.1239	0.1698	0.2097	0.1152	0.2159
3	0.1224	0.1124	0.1370	0.1201	0.1102	0.1259
6	0.1113	0.1230	0.1149	0.1119	0.1152	0.1123

A more rigorous comparison can be drawn between the SVAR and BSVAR models. As shown in Tables 6 and 8, both models exhibit comparable forecasting performance across all horizons. While the BSVAR framework offers theoretical advantages—such as shrinkage via regularization, robustness in small samples, and the incorporation of prior beliefs—it does not consistently outperform the SVAR model in predictive accuracy. This outcome likely reflects a tradeoff: while Bayesian models can improve short-term forecasts through regularization, their reliance on priors and additional parameter uncertainty can hinder performance over longer horizons, especially when priors are not well-aligned with the true data-generating process.

Notably, in Scenarios 1 and 3, the non-recursive Bayesian specification excels at one-step-ahead forecasts, but the frequentist recursive SVAR performs better at medium to long horizons. The recursive structure likely contributes to its stability, reducing overfitting and improving reliability as forecast errors accumulate. In summary, while the Bayesian approach enhances short-term flexibility, the recursive SVAR model proves more robust for long-term forecasting, highlighting a classic bias-variance tradeoff in empirical model performance.

4.5 Dynamic Factor Model Augmentation

4.5.1 Principal Components

The study first determines the number of latent factors and the appropriate lag structure to estimate the Dynamic Factor Model (DFM). These are selected to balance model fit with parsimony: enough principal components (PCs) are retained to explain a substantial share of the total variance in the data without risking overfitting. Specifically, the paper sets a discretionary threshold of 50% cumulative variance explained, beyond which additional factors yield diminishing returns in explanatory power (see Appendix L). Based on this criterion, the study retains the first three PCs. Consequently, the information criteria are evaluated to determine the appropriate number of lags in the factor transition equation. The Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) consistently suggest choosing two lags, which will be adopted for all subsequent estimations.

The PCs are computed for the monthly data series based on their correlation matrix. The first PC explains 20.3% of the individual variances of the total variance, with the second and third

PCs accounting for 15.2% and 12.3% respectively. Together, they explain almost 50% of the total variance. The "individual" variances of each series explained by the different PCs are as follows.

Table 9: Proportion of Variance in Each Variable Explained by PC1–PC3

Variables	PC1	PC2	PC3
Brent Crude Oil	0.472	0.187	0.026
Gas Prices	0.134	0.062	0.013
Industrial Production	0.070	0.289	0.061
Labour Cost Index	0.152	0.020	0.017
Euro-area inflation rate	0.022	0.057	0.573
GECON Index	0.716	0.014	0.022
Exchange rate EUR/USD	0.166	0.100	0.354
Interest-rate differential	0.049	0.446	0.149
Real disposable income	0.143	0.007	0.032
Caldara–Iacoviello Index	0.105	0.313	0.031

PC1 captures global supply-side pressures, especially from energy markets and international economic momentum. The high loading for Brent and GECON suggests it reflects conditions in global commodity markets and broader macroeconomic activity. A weaker euro, via the exchange rate, often corresponds to rising import prices, linking this PC to cost-push inflationary forces. PC2 represents monetary policy and demand-driven forces. Interest rate differentials strongly influence capital flows and domestic demand. Industrial production aligns with real activity, and geopolitical uncertainty can impact investment and sentiment, which shape demand-side inflation dynamics. PC3 is likely capturing idiosyncratic inflation movements influenced by domestic and external macro-financial conditions. The dominant share of inflation variance and moderate contributions from monetary variables suggest that this PC reflects persistent inflation drivers not entirely captured by global or cyclical variables.

This initial result aligns with the theoretical background in the Introduction, where supply forces were expected to be the main drivers of inflation.

4.5.2 Forecasting

Figure 7 illustrates the one-step-ahead out-of-sample forecast performance of the Dynamic Factor Model (DFM) for euro-area inflation. The figure plots actual inflation growth rates against DFM predictions over the test period, using a rolling-window estimation scheme based on pre-2023 data. At each forecast step, the model is re-estimated using the most recent 95 months of available data, allowing it to adapt to evolving macroeconomic dynamics. Forecasts are then generated conditional on information available up to time $t - 1$. The DFM demonstrates a reasonable ability to capture the directional movements and local turning points in inflation dynamics, although the magnitude of

some forecast errors is non-negligible. Additionally, it should be noted that the model underperforms from August to November. This showcases the difficulty of forecasting when summarizing a broad set of macroeconomic variables through latent factors.

Next, the paper assesses the DFM’s quantitative forecast precision via its out-of-sample Root Mean Squared Errors (RMSEs). Over the 12 months, the one-step-ahead, three-step-ahead, and six-step-ahead RMSEs are 0.0659, 0.0644, and 0.0684, respectively. These are the lowest errors observed across all competing SVAR and BSVAR competitions and confirm DFM’s capacity to maintain accuracy even as forecasting horizons grow. This makes the DFM a reliable framework for real-time policy-relevant forecasting.

Comparison to the Bayesian SVAR

Although other scenarios exhibit lower forecasting accuracy compared to the dynamic factor model, a key question arises: Is the observed improvement in performance truly attributable to the dynamic factor model itself, or simply the result of incorporating additional variables? Scenario 4—which includes all ten variables—is evaluated within a Bayesian framework to investigate this.

At first glance, Figure 8 may suggest that the Bayesian model is better. However, the series is plotted on inconsistent scales, which distorts the comparison. Once the axes are standardized, the Bayesian specification falls short of the DFM.

This conclusion is reinforced by the RMSE values, which are consistently higher for the Bayesian model across all forecast horizons: one-step-ahead (0.1320), three-step-ahead (0.1403), and six-step-ahead (0.1339). While including more variables may improve performance relative to simpler Bayesian scenarios in the short run, it does not outperform the DFM. These results emphasize the risks of overfitting in high-dimensional settings and highlight the DFM’s advantage in addressing this challenge.

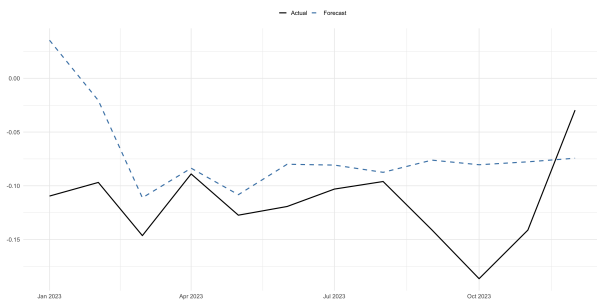


Figure 7: Actual Inflation vs. DFM One Step Ahead Forecast from Jan 2023 to Dec 2023

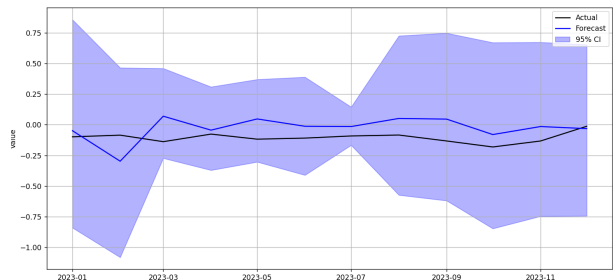


Figure 8: Actual Inflation vs. Bayesian One Step Ahead Forecast from Jan 2023 to Dec 2023

4.5.3 Evaluation of Impulse Responses

The paper examines inflation’s impulse response functions (IRFs) to orthogonalized shocks in the latent factors to understand better the dynamic transmission mechanisms captured by the estimated DFM. These factors summarize co-movements across the macroeconomic dataset and

can be interpreted as capturing broad underlying structural dynamics. The IRFs are derived from the state-space representation of the estimated DFM, where structural shocks are identified via a Cholesky decomposition. Each line in the plot shows the estimated path of inflation following a one-unit orthogonalized shock to a single factor, holding all else constant. The study can infer how inflation dynamics are primarily driven by analyzing these responses' shape, magnitude, and persistence over a 12-month horizon.

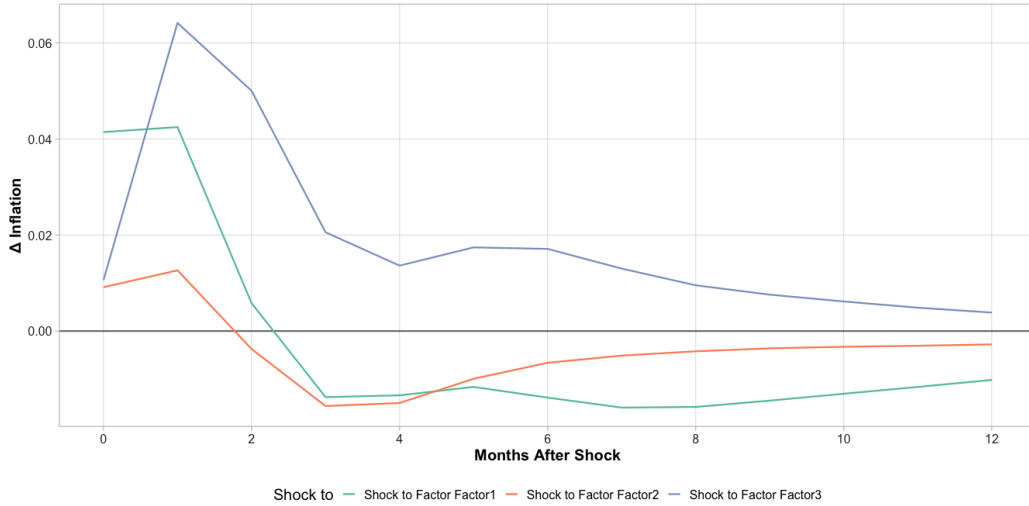


Figure 9: Inflation response to factor shocks for DFM over a 12-month horizon.

The response of inflation to a Factor 1 shock is immediate and positive in the short term, peaking at around month 1 and then turning negative around month 3, yet ending slightly positive by the end. This pattern is consistent with global or cyclical factors, such as a commodity price or activity shock that pushes inflation up and cools off as growth slows and the economy adapts. Given that this factor explains global supply-side pressures, this IRF can be interpreted as responses to supply-side cost-push pressures, such as shocks to commodity prices (e.g., oil, gas) and global economic activity. These raise input costs quickly and pass through to consumer prices, but their effects diminish as supply chains and inventories adjust.

Factor 2 is the smallest and least persistent factor. Factor 2 shows a more modest and transitory response. Inflation rises slightly initially, dips into negative territory around months 3–4, and then slowly returns to baseline by month 12. This pattern is typical for transitory effects—like an increase in demand—that push prices up sporadically before returning to their long-run equilibrium. Given the interpretation of the factor as monetary policy and demand-driven force, this IRF reflects the impact of monetary tightening or domestic demand disturbances, factors that influence inflation via real activity and credit conditions but whose effects are more limited and possibly offset over time.

Factor 3 path is the largest, seems unidirectional—constantly positively impacting inflation, and is slow to mean-revert. The IRF of this factor confirms the initial interpretation of the factor as a domestic idiosyncratic component. Indeed, the shock to Factor 3 is distinct from the other two: no overshoot or sign change and keeps inflation above zero for the whole year. That independence and

persistence are signs of an idiosyncratic—or "third" factor driving inflation. Moreover, the effect is persistent and dominant: At the impact period (month 0–1), the Factor 3 shock lifts Δ inflation by ≈ 0.06 ; even after 6–12 months, it is still the largest contributor.

5 Conclusion

The paper’s ultimate question is, "How can improved VAR models better forecast the effect and persistence of different shocks on Eurozone inflation?" The evidence suggests that no specification consistently excels across all time horizons; different models outperform each other when considering different time frames. The results suggest that inflation is ever-changing, and different policy goals require different specifications.

A demand-only recursive SVAR is the optimal tool when interested in short-term inflation dynamics. The results confirm that short-term inflation dynamics are primarily driven by real disposable income and the EUR/USD exchange rate, as these variables capture the volatile but anticipated response of prices to shifts in spending power and import costs. The RMSE of the demand-only model is roughly 55 % smaller than the full model at the one-step horizon (0.094 vs 0.204) and remains the best performer out to three months ahead. Because demand shocks peak quickly and dissipate within a quarter, adding supply variables at this stage only introduces noise.

Beyond three to six months, however, the dynamics change. A broader supply-and-demand SVAR is required to prevent systematic under-prediction. Cost-push shocks from Brent oil and gas prices have minor initial effects yet a significant prolonged effect. This result indicates that shocks in the energy sector have a persistent effect on inflation. Moreover, leaving the supply effects out fails to capture the inflation persistence, as reflected in the worse forecast accuracy of demand-only models. In other words, supply variables gradually increase their magnitude in medium-term inflation dynamics, requiring more attention from policymakers to avoid periods of sustained inflation.

Incorporating Bayesian shrinkage via a BSVAR with Minnesota priors on top of the SVAR structure provides valuable insights, but a closer comparison between the SVAR and BSVAR models reveals key differences in forecasting performance. In particular, while the non-recursive Bayesian specification excels in short-term forecasts, the recursive SVAR model proves more reliable over medium to long horizons. The stability of the recursive structure helps reduce overfitting and improves forecast accuracy as errors accumulate. While the Bayesian approach provides flexibility and enhanced inference, the recursive SVAR model is more robust for long-term forecasting, highlighting the classic bias-variance tradeoff in model performance.

Finally, the variance in the dataset is condensed into three latent factors through DFAVAR. Global energy supply and activity shocks dominate the first factor; the second captures monetary policy and demand-cycle movements; and the third factor represents a persistent domestic inflation core not explained by the other forces, suggesting a constant idiosyncratic component in inflation. The approach identifies turning points well and helps avoid omitted variable bias. However, it also fails to forecast inflation in volatile periods—such as the energy-price spikes in mid-2023—suggesting

a changing regime in inflation. Nevertheless, the DFAVAR consistently delivers the most accurate forecasts—exhibiting the lowest RMSE across every scenario. Its factor-based structure balances complexity and parsimony, mitigating overfitting and improving robustness in medium-term inflation projections.

However, the research presents limitations that could restrict the results’ scope but could be built upon for further research. Indeed, the analysis relies on a relatively brief monthly span from January 2015 to January 2024, which may not enclose different inflation regimes or structural breaks. Moreover, several intuitively important series (GDP, unemployment, GSCPI) were excluded for non-stationarity, and quarterly variables were Kalman-smoothed to generate monthly data—potentially introducing measurement error and bias into impulse-response estimates. Finally, forecast performance is evaluated solely over the volatile post-pandemic year 2023, so its generalisability to calmer periods or different policy regimes remains untested.

To build on the VAR-based exploration of Eurozone inflation, future work could include non-linear and machine-learning methods—such as regime-switching random forests or LSTM networks—on top of the SVAR/BSVAR residuals to detect turning points in supply-chain pressure indices (GSCPI, gas, Brent) that the linear models may smooth over. Similarly, implementing the SVAR, BSVAR and DFAVAR frameworks in a real-time data-release environment (feeding in first-release data or energy-price vintages) would reveal how forecast accuracy degrades under informational frictions. Finally, expanding beyond the single Eurozone block to a cross-country panel of member states, each with its own inflation and policy reaction function, would test whether the factor-augmented dashboard and recursive identification hold in economies with different exposure to external shocks.

Appendix

A Overview of inflation and its key drivers

Variables	Transformation	Frequency	Source
Euro area inflation rate	Growth rate	Monthly	Rate inflation
Brent Crude Oil Prices	Log-difference	Monthly	Yahoo Finance
Gas Prices	Log-difference	Monthly	FRED
Exchange rate EUR/USD	Log-difference	Monthly	ECB
Disposable income	Log-difference	Monthly	FRED
Global Economic Conditions (GECON) index	no transformation	Monthly	Baumeister et al. (2022)
Industrial production (IP)	Log-difference	Monthly	Eurostat
Gross Domestic Product (GDP)	Kalman-filtering and Log-difference	Quarterly	Eurostat
Labour Cost Index (LCI)	Kalman-filtering and Log-difference	Quarterly	Eurostat
Unemployment rate	First difference	Monthly	Eurostat
Global Supply Chain Pressure Index (GSCPI)	First difference	Monthly	NY Fed
Interest rate	Log-difference	Monthly	FRED
Electricity prices	Kalman-filtering and Log-difference	Quarterly	Ember
Caldara-Iacoviello Index	no transformation	Monthly	Matteo Iacoviello

Table 10: Overview of inflation and its key drivers

Note: This table summarizes the variables included in the analysis, along with their transformations, frequencies, and data sources. The baseline specification—motivated by Bańbura et al. (2023)—includes Brent Crude Oil Prices, Gas Prices, the Euro area inflation rate, the EUR/USD exchange rate, and disposable income. The selection is grounded in both theoretical considerations and empirical evidence. Most variables have been transformed to achieve stationarity and harmonized to a monthly frequency. Standardization is applied to ensure comparability across series.

B Theorem and Proof of the Moving Average Representation

Given a stable VAR(p) process of the form:

$$Y_t = A_1 Y_{t-1} + A_2 Y_{t-2} + \cdots + A_p Y_{t-p} + e_t, \quad (16)$$

if the roots of the characteristic polynomial $\det(A(L)) = 0$ lie outside the unit circle, then the process can be written as a moving average (MA) process of infinite order:

$$Y_t = \sum_{j=0}^{\infty} C_j e_{t-j}. \quad (17)$$

Proof Sketch. The $\text{VAR}(p)$ process can be expressed in lag operator form as:

$$A(L)Y_t = e_t, \quad (18)$$

where $A(L) = I - A_1L - A_2L^2 - \dots - A_pL^p$ is the lag polynomial. Under the stability condition that the roots of the characteristic equation $\det(A(L)) = 0$ lie outside the unit circle, the lag polynomial $A(L)$ is invertible. By the Neumann series (see Appendix C), we can write the inverse of $A(L)$ as:

$$A(L)^{-1} = \sum_{j=0}^{\infty} C_j L^j, \quad (19)$$

where the series converges, and therefore:

$$Y_t = A(L)^{-1}e_t = \sum_{j=0}^{\infty} C_j e_{t-j}. \quad (20)$$

C Neumann Series and Invertibility of Lag Polynomials

Theorem (Neumann Series). Suppose that T is a bounded linear operator on a normed vector space X . If the Neumann series converges in the operator norm, then $I - T$ is invertible and its inverse is given by:

$$(I - T)^{-1} = \sum_{k=0}^{\infty} T^k, \quad (21)$$

where I is the identity operator in X .

Proof Sketch. Consider the partial sums:

$$S_n := \sum_{k=0}^n T^k. \quad (22)$$

Then:

$$(I - T)S_n = (I - T) \sum_{k=0}^n T^k \quad (23)$$

$$= \sum_{k=0}^n T^k - \sum_{k=0}^n T^{k+1} \quad (24)$$

$$= I - T^{n+1}. \quad (25)$$

Taking the limit as $n \rightarrow \infty$ and assuming that $\|T\| < 1$, the following result is obtained:

$$\lim_{n \rightarrow \infty} (I - T)S_n = I. \quad (26)$$

Thus, $(I - T)$ is invertible and:

$$(I - T)^{-1} = \sum_{k=0}^{\infty} T^k. \quad (27)$$

This result is analogous to the geometric series in R . In the context of the VAR(p) process, T corresponds to the lag polynomial operator $A(L)$, and the invertibility condition corresponds to the stability of the process (all eigenvalues outside the unit circle).

D Derivations Variance - Structure Minnesota Priors

The variance structure is specified as follows:

$$V_{i,kk} = \begin{cases} \frac{\lambda_1}{l^2}, & \text{for the coefficient on the } l\text{-th lag of variable } i \text{ (own lag)} \\ \frac{\lambda_2 s_i^2}{l^2 s_j^2}, & \text{for the coefficient on the } l\text{-th lag of variable } j, j \neq i \text{ (cross-lag)}, \end{cases}$$

where λ_1, λ_2 and λ_3 are shrinkage parameters controlling prior belief strength. Additionally, s_i^2 and s_j^2 are the sample variances of residuals of variables i and j , respectively, estimated using a standard VAR model.

The term $\frac{1}{l^2}$ of the variance matrix $V_{i,kk}$ ensures that the coefficients in the higher lags shrink more, reducing the risk of overfitting the distant lag relationships. For own lags, the shrinkage becomes stronger as the lag order increases:

$$V_{i,kk} \propto \frac{1}{l^2}.$$

Shrinkage works by pulling the parameter estimates toward the prior mean, which results in a penalty for large coefficients unless the data strongly suggests they are important. For cross-lagged terms, the shrinkage depends on the relative variability of the variables. If s_i^2 is much larger than s_j^2 , the model will shrink the cross-lagged coefficients more, reflecting that the relationship between the variables may not be as strong. For the intercept, the shrinkage depends on the variability of the variable itself, ensuring that intercepts for more variable series receive less shrinkage.

E Derivations - Recursive Bayesian VAR

The distribution first generates a seed to obtain the correlation matrix R . The shape parameter η controls the concentration of the distribution around the identity matrix, ensuring valid correlation structures. The probability density function for a $k \times k$ matrix R is:

$$p(R; \eta) = C \times [\det(R)]^{\eta-1}, \quad (28)$$

where C is the normalizing constant, given by:

$$C = 2^{\sum_{k=1}^{d-1} (2\eta - 2 + d - k)(d - k)} \prod_{k=1}^{d-1} \left[B \left(\eta + \frac{d - k - 1}{2}, \eta + \frac{d - k - 1}{2} \right) \right]^{d - k}. \quad (29)$$

When $\eta = 1$, the distribution is uniform over the space of all correlation matrices, meaning it is uniform over the set of positive definite matrices with a unit diagonal.

F Likelihood Function and Posterior Distribution

The likelihood of function of the data is written as:

$$L(\theta | Y) = \prod_{t=1}^T N(Y_t | X_t \beta, \Sigma_\epsilon) = \prod_{t=1}^T \frac{1}{(2\pi)^{k/2} |\Sigma_\epsilon|^{1/2}} \exp \left(-\frac{1}{2} \epsilon_t^T \Sigma_\epsilon^{-1} \epsilon_t \right), \quad (30)$$

where X_t is the matrix of lagged variables $Y_{t-1}, Y_{t-2}, \dots, Y_{t-p}$, $\beta = [B_1, B_2, \dots, B_p]$ is the coefficient matrix of the model, and θ represents the set of all model parameters $(B_1, B_2, \dots, B_p, D^{-1}, \Sigma_\epsilon)$. The dimension of Y_t is k .

And the posterior distribution is given by:

$$p(\theta | Y) \propto \prod_{t=1}^T N(Y_t | X_t \beta, \Sigma_\epsilon) \cdot \prod_{i=1}^p N(B_i | 0, \Sigma_B) \cdot IW(D^{-1} | \Psi_0, \nu_0) \cdot IW(\Sigma_\epsilon | \Psi_0, \nu_0). \quad (31)$$

G Standard VAR results

Table 11: Forecast RMSE for Scenario 1–3 using Standard VAR

Horizon	Scenario 1	Scenario 2	Scenario 3
1	0.2170	0.0999	0.2241
3	0.0889	0.0866	0.0914
6	0.0947	0.0951	0.0943

H Derivation of Long-run restrictions in SVAR

Assume for simplicity that:

$$\tilde{C}_j = \begin{bmatrix} \tilde{c}_j^{(s)} \\ \tilde{c}_j^{(d)} \end{bmatrix}, \quad (32)$$

where $\tilde{c}_j^{(s)}$ is the coefficient associated with the structural shocks on the supply side, and $\tilde{c}_j^{(d)}$ is the coefficient associated with the structural shocks on the demand side, in the moving average

representation at time j .

Consider the impulse response of a variable in Y_t , in this case $\Delta\pi$, to the structural shocks represented by:

$$\begin{bmatrix} \epsilon_{t-j}^s \\ \epsilon_{t-j}^d \end{bmatrix}.$$

The following is known:

$$\pi_t = \pi_{t-1} + \Delta\pi_t = \pi_{t-j} + \sum_{i=0}^{j-1} \Delta\pi_{t-i}, \quad (33)$$

so:

$$\frac{\partial \pi_t}{\partial \begin{bmatrix} \epsilon_{t-j}^s \\ \epsilon_{t-j}^d \end{bmatrix}} = \sum_{i=1}^j \tilde{c}_{i,p}^{(d)}. \quad (34)$$

$$\rightarrow \sum_{i=1}^{\infty} \tilde{c}_{i,p}^{(d)} \quad (35)$$

where p is the index of inflation in the demand side vector of coefficients.

The paper considers the following identification matrix:

$$\sum_{i=1}^{\infty} \tilde{c}_i = \begin{pmatrix} j_{11} & j_{12} & j_{13} & 0 & 0 \\ j_{21} & j_{22} & j_{23} & 0 & 0 \\ j_{31} & j_{32} & j_{33} & j_{34} & j_{35} \\ j_{41} & j_{42} & j_{43} & j_{44} & j_{45} \\ j_{51} & j_{52} & j_{53} & j_{54} & j_{55} \end{pmatrix}, \quad (36)$$

based on the assumption that demand-side variables do not affect supply-side dynamics in the long run. Given the focus on disentangling cost-push and demand-pull drivers of inflation, the study allows both types of shocks to exert persistent effects on inflation in the long-run. Based on equation 5, the following holds true:

$$\sum_{i=1}^{\infty} \tilde{C}_j = \left(\sum_{i=1}^{\infty} C_j \right) D = C(1)D = A(1)^{-1}D, \quad \text{where } C(L) = \sum_j C_j L^j = A(L)^{-1}.$$

The following steps are considered to identify the impact of the structural shocks incorporating the long-run restrictions, i.e. the D matrix:

Step 1 Run OLS regression $Y_t = A_1 Y_{t-1} + \dots + A_p Y_{t-p} + e_t$, get OLS coefficients $A_1, \dots, A_p$ and variance of the residuals Ω

Step 2 Find D such that

$$\left(I_k - \sum_{j=1}^p A_j\right)^{-1} D = \begin{pmatrix} * & * & * & 0 & 0 \\ * & * & * & 0 & 0 \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{pmatrix} \quad \text{and} \quad \Omega = DD^\top.$$

I Optimized structural matrix and long-run impact matrix of Scenarios 2 and 3

Table 12: Structural D Matrix for Scenario 2

	$\%\Delta\pi$	$\%\Delta Y_{dis}$	$\%\Delta E$
$\%\Delta\pi$	1.0075	0.0000	0.0000
$\%\Delta Y_{dis}$	-0.0469	0.8277	0.0000
$\%\Delta E$	0.2226	-0.1929	0.9128

Table 13: Long-Run Impact Matrix $C(\infty) \cdot D$ for Scenario 2

	$\%\Delta\pi$	$\%\Delta Y_{dis}$	$\%\Delta E$
$\%\Delta\pi$	0.9534	-0.0042	-0.0681
$\%\Delta Y_{dis}$	-0.0261	0.5226	0.0342
$\%\Delta E$	0.3157	-0.2450	1.2987

Table 14: Structural D Matrix for Scenario 3

	π_{Brent}	π_{gas}	$\%\Delta\pi$	$GECON$	$\%\Delta i$	$\%\Delta Y_{dis}$	$\%\Delta E$
π_{Brent}	0.8994	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
π_{gas}	-0.0000	0.9348	0.0000	0.0000	0.0000	0.0000	0.0000
$\%\Delta\pi$	0.0705	-0.0000	0.9842	0.0000	0.0000	0.0000	0.0000
$GECON$	0.4608	0.0733	-0.0713	0.5673	0.0000	0.0000	0.0000
$\%\Delta i$	0.0005	-0.1856	0.3073	-0.0003	0.8751	0.0000	0.0000
$\%\Delta Y_{dis}$	-0.1092	-0.0316	-0.0195	-0.0000	-0.0372	0.7136	0.0000
$\%\Delta E$	0.1069	0.0636	0.1778	0.1228	0.0464	-0.1566	0.8121

Table 15: Long-Run Impact Matrix $C(\infty) \cdot D$ for Scenario 3

	π_{Brent}	π_{gas}	$\% \Delta \pi$	<i>GECON</i>	$\% \Delta i$	$\% \Delta Y_{d,US}$	$\% \Delta E_{\text{EUR-USD}}$
π_{Brent}	1.2201	-0.0828	-0.0245	-0.1262	0.1219	-0.0261	0.2313
π_{gas}	0.4806	1.0128	-0.0744	0.1986	-0.1283	-0.0230	0.2076
$\% \Delta \pi$	0.1583	-0.1418	0.9153	-0.0334	-0.0290	0.0043	-0.0554
<i>GECON</i>	1.3723	0.0797	-0.1631	0.7789	-0.1002	0.0826	0.2103
$\% \Delta i$	0.0829	0.0312	0.5298	-0.0653	1.1479	-0.0322	-0.1037
$\% \Delta Y_{d,US}$	-0.3902	-0.0558	0.0349	-0.1924	-0.0146	0.3962	0.0177
$\% \Delta E_{\text{EUR-USD}}$	0.0784	-0.3120	0.1341	0.3138	-0.3640	-0.1340	1.0901

J Bayesian structural matrices Scenario 2-3

Table 16: Point estimates for structural matrix D (mean of posterior) — Scenario 2

	$\% \Delta \pi$	$\% \Delta Y_{dis}$	$\% \Delta E$
$\% \Delta \pi$	0.1076	0.9371	0.4075
$\% \Delta Y_{dis}$	-1.1290	-0.0069	-0.6174
$\% \Delta E$	-0.7515	-0.6992	0.3398

Table 17: Point estimates for structural matrix D (mean of posterior) — Scenario 3

	π_{Brent}	π_{gas}	$\% \Delta \pi$	<i>GECON</i>	$\% \Delta i$	$\% \Delta Y_{dis}$	$\% \Delta E$
π_{Brent}	-0.5291	0.0584	-0.1576	0.3913	0.1422	-0.0497	-0.0072
π_{gas}	0.0293	-0.3648	-0.2943	0.0149	0.0391	0.0054	-0.0456
$\Delta \pi$	0.4345	0.4412	0.1870	0.4162	0.4681	0.4265	0.3851
<i>GECON</i>	0.3764	0.0431	-0.0828	-0.7399	-0.0355	0.0609	-0.0701
$\% \Delta i$	-0.1097	-0.1972	-0.4252	-0.2853	-0.4299	-0.2038	-0.2785
$\% \Delta Y_{dis}$	0.0321	0.0340	-0.2548	-0.0349	0.0982	-0.5493	0.0307
$\% \Delta E$	-0.1621	-0.1884	-0.0652	-0.1205	-0.1755	-0.2445	0.2902

K IRFs to structural shocks Bayesian Scenario 2-3

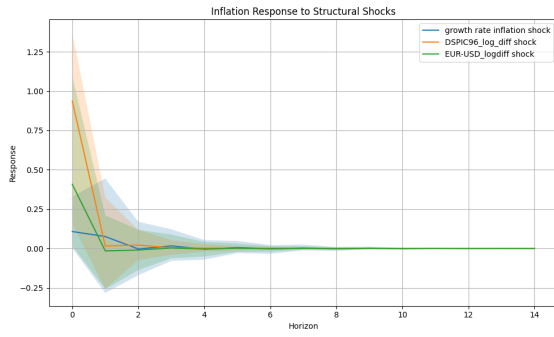


Figure 10: Impulse responses of Scenario 2 using non-recursive ordering

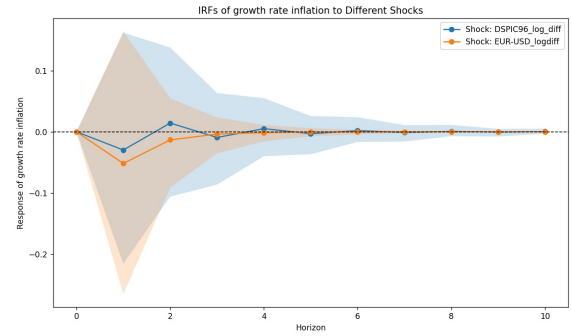


Figure 11: Impulse responses of Scenario 2 using recursive ordering

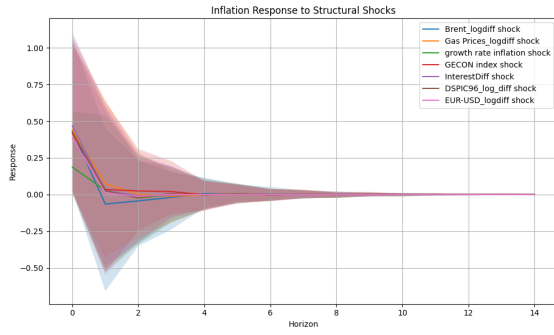


Figure 12: Impulse responses of Scenario 3 using non-recursive ordering

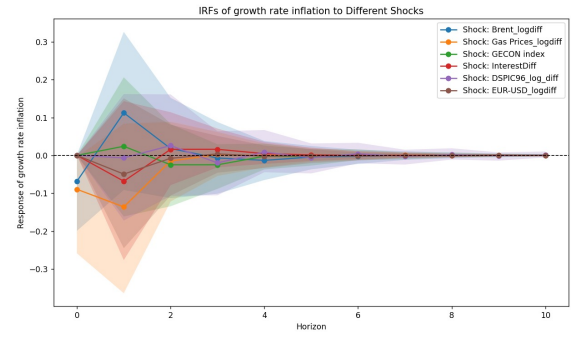


Figure 13: Impulse responses of Scenario 3 using recursive ordering

L Principal Component Analysis of FAVAR

Table 18: Principal Component Analysis: Eigenvalues and Variance Explained

Principal Component	Eigenvalue	Proportion of Variance	Cumulative Variance
PC1	2.030	0.203	0.203
PC2	1.517	0.157	0.355
PC3	1.278	0.128	0.483
PC4	1.141	0.114	0.597
PC5	0.938	0.094	0.691
PC6	0.843	0.084	0.775
PC7	0.716	0.072	0.846
PC8	0.625	0.063	0.909
PC9	0.564	0.056	0.965
PC10	0.348	0.035	1.000

M Correlation matrix of variables

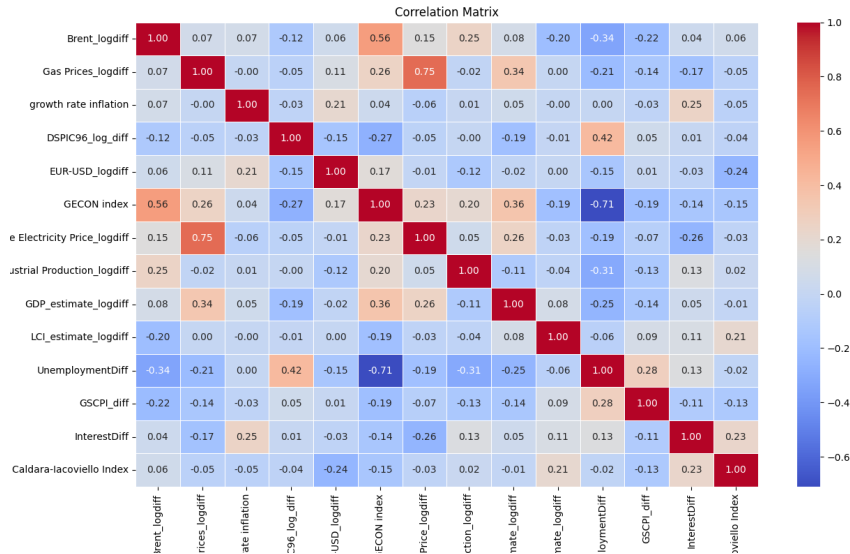


Figure 14: Correlation Matrix of all variables

N Transformation into a VAR representation in FAVAR model

To derive a reduced-form representation, equation 12 must be substituted into equation 11:

$$(I - \delta_i(L)L)y_{i,t} = \lambda_i(L)f_t + w_{i,t} \quad \text{where } \lambda_i(L) = (I - \delta_i(L)L)\tilde{\lambda}_i(L).$$

This equation provides an alternative specification of the model. The system can also be written in matrix form, where $D(L)$ is a block diagonal matrix capturing the dynamics of the idiosyncratic components.

$$Y_t = D(L)Y_{t-1} + \lambda_i(L)f_t + w_t \quad \text{where } D(L) = \begin{bmatrix} \delta_1(L) & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \delta_n(L) \end{bmatrix}.$$

The model can also be rewritten in static form by introducing an augmented factor vector:

$$\begin{aligned} Y_t &= D(L)Y_{t-1} + \Lambda F_t + w_t \\ F_t &= \Phi(L)F_{t-1} + G\eta_t, \end{aligned} \quad \text{where } F_t = [f_{t-1}, \dots, f_{t-p}].$$

Finally, everything can be written together as a VAR:

$$\begin{bmatrix} F_t \\ Y_t \end{bmatrix} = \begin{bmatrix} \Phi(L) & 0 \\ \Lambda\Phi(L) & D(L) \end{bmatrix} \begin{bmatrix} F_{t-1} \\ Y_{t-1} \end{bmatrix} + \begin{bmatrix} I & 0 \\ \Lambda & I \end{bmatrix} \varepsilon_t, \quad \varepsilon_t = G\eta_t + w_t. \quad (37)$$

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